

V_us reproduced BLIND (0.31%) → DERIVED — the quark sector’s first forward MIXING prediction, from the same SVD that gave m_s/m_d=20; flagship CLEARS both audit gates (Cal §135 re-verify + Keeper), Casey GO (one non-blocking C₂ polish); registry synced T2527–T2529

Keeper

2026-07-29

K994 — The first quark prediction lands blind; the flagship is GO-clean

★★ V_us reproduced BLIND (0.31%) → DERIVED — the first quark MIXING prediction

Elie toy 4915: $V_{us} = \sqrt{m_d/m_s} = 1/\sqrt{20} = 1/(\text{rank} \cdot \sqrt{n_C}) = 0.2236$ vs obs 0.2243 → 0.31%. Three things make it a genuine FORWARD reproduction, not a postdiction: - Derived from the SVD, not imported: diagonalizing the down-sector texture-zero gives $\tan \theta_C = \sqrt{m_d/m_s}$ as the angular part — the Gatto relation IS the geometry (F689), not an external input. - Only the forced ratio 20 went in — never the observed V_us, never observed masses. Clears the exact blind bar the tau’s 71 failed. - Honest reporting: Elie gave both the leading (0.31%) and the exact 2-gen (2.71%) values, not just the one closest to data. The up-sector correction between them is Tier-2. - RULING: V_us leading → DERIVED (Derived-calc from the DERIVED m_s/m_d=20, via the DERIVED Gatto/down-sector-diagonalization relation; blind, 0.31%). The quark sector’s first mixing angle at Derived, AND a forward prediction. The up-sector residual (0.31% gap) is honestly Tier-2. This validates the addresses are pinned right twice over (m_s/m_d=20 AND V_us).

★ Quark scorecard (all fell out of the $\mathbb{C}^3$ SVD, none aimed at)

m_s/m_d = 20 DERIVED · V_us blind 0.31% DERIVED · anchor $3\pi/128$ EXACT · top-ceiling m_t ≤ 174.1 DERIVED · everything else honestly Tier-2, over-fits rejected. Three Derived quark quantities, two of them (m_s/m_d, V_us) forward/blind. The colored sector is on solid ground — the restraint (refusing to calibrate) is what makes the forward reproductions mean something.

★ FLAGSHIP — clears BOTH audit gates → Casey GO

- Cal §135 re-verify: CLEARS. All 5 fixes landed; muon never-bare confirmed (abstract/§3/§6 all carry “Derived (e=n without counterexample)”); the only bare-“Derived” hits are electron + tau-home, both correctly Derived); count/tau at ratified tiers; only “pending” left = “pending Casey’s GO.”
- Keeper: CLEARS (tiers==record; $6\pi^5$ re-audited K992; Value-3 double-verified K991; ledger ship-consistent after Grace’s base-tier blocker fix).
- One NON-BLOCKING polish (Cal): the §6 electron-row caveat reads “ $N_c! \cdot \pi^{n_C}$ ” while the abstract/§3 canonicalize the 6 as C₂ (K992, the Bergman-Laplacian spectral gap). Both = 6, but two presentations invite the “6 is easy to make” reading the muon’s 24 fended off — the table should cite C₂, one canonical meaning. A one-cell Lyra edit; tier (Derived, target-innocent) correct either way; worth doing before a headline paper, does NOT gate GO.

- STATUS: GO-clean. Casey's call, repo-internal, not pushed. Recommend the one C_2 edit first (cheap, avoids a fended-off reading).

★ Registry sync (Grace flagged — Keeper's lane)

Grace built `play/verify_ac_graph.py` (self-checking, checksum) and registered this week's results as graph nodes: T2527 (three generations forced), T2528 ($\tau =$ boundary mode), T2529 (quark $m_s/m_d=20$). Keeper syncs the theorem REGISTRY (`notes/BST_AC_Theorem_Registry.md`) to match — add T2527–T2529 with their tiers (T2527 DERIVED/Value-3; T2528 DERIVED-home; T2529 DERIVED) + a T2530 candidate slot for V_{us} -blind if it registers. (Registry-graph consistency is now checkable via Grace's verifier.)

What's next (quark predictions — the falsifiable payoff)

V_{us} passed; the predictions we never used are next: V_{cb} , m_c/m_u , and the neutrino Δm^2 — the up-type ratio-of-ratios (Tier-2), which need Lyra's forced Pochhammer arguments. Elie runs all three BLIND the moment Lyra pins them; a forward number that lands there is where the colored sector earns its keep. Keeper rules each against the blind bar; a calibrated prediction = rejected.

— K994, Keeper, 2026-07-29. V_{us} blind 0.31% → Derived (first quark mixing prediction, forward); flagship GO-clean both gates (one non-blocking C_2 polish); registry T2527–T2529 to sync. Next: predict $V_{cb}/m_c/m_u/\Delta m^2$ blind. See `[[Keeper_K993_QUARK_SVD_first_results_m_s_over_m_d_20_DERIVED_first_quark_mass_ratio_anchor_3pi128_e07-29]]`, Elie toy 4915, Lyra F689.